OAKLAND COMMUNITY COLLEGE®

ROBOTICS/AUTOMATED SYSTEMS TECHNOLOGY

ROB 2140
Siemens FUNCTION BLOCK LABS

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IMPORTANT FOR GRADE

For each lab, the **named Watch Window** must be created and displayed during program execution for credit.

See graphic Completed Watch Window for required tags.

TABLE OF CONTENTS

SIEMENS FUNCTION BLOCK LABS

LAB: SS1
PROCESS MODEL: SS1 *{last name} {first name}*
OBJECTIVE: Create a simple Stop-Start circuit with function block instructions.
SOFTWARE: Siemens TIA

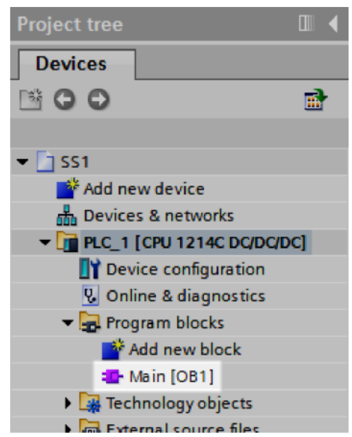
START - STOP circuit for a single motor control with ON - OFF indicators.

PROGRAM LOCATION:
Siemens Folder
Course Day/Time Folder
Languages Folder
Function Block Folder

FOLDER\FILE NAME: SS1

PROGRAM BLOCK:

Expand the Project Tree and select the routine **Main (OB1)**.



The blank editor will be displayed.

PLC Tags

INPUT:

Address	PLC IO Tag	Description
I0.0	START	Pushbutton Momentary N.O.
I0.1	STOP	Pushbutton Momentary N.C.

OUTPUT:

Address	PLC IO Tag	Description
Q0.0	PL1_OFF	Pilot Light 1 OFF (Red)
Q0.1	PL2_ON	Pilot Light 1 ON (Green)
Q0.2	MOTOR	Motor Control

FUNCTION BLOCK ROUTINE

See Function Block Assist for program.

Add the following instructions.

Instruction Folder for **Bit logic operations**. Bit logic operations

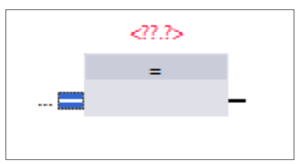
>=1 OR logic operation

& AND logic operation

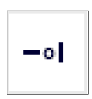
-[=] Assignment

NOT - Normally Closed

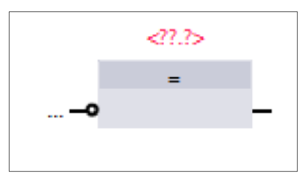
Select the Input.



Select the Invert RLO icon.



The *Results of Logic Operations* is equivalent of a Normally Closed input.

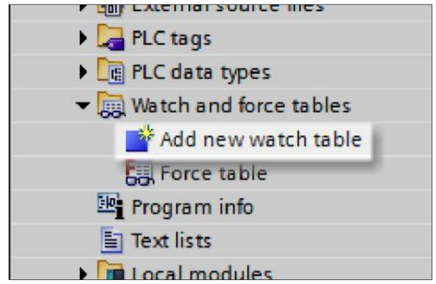


continue

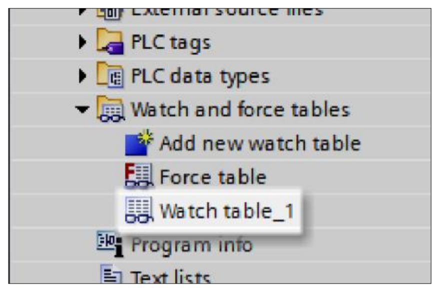
WATCH WINDOW

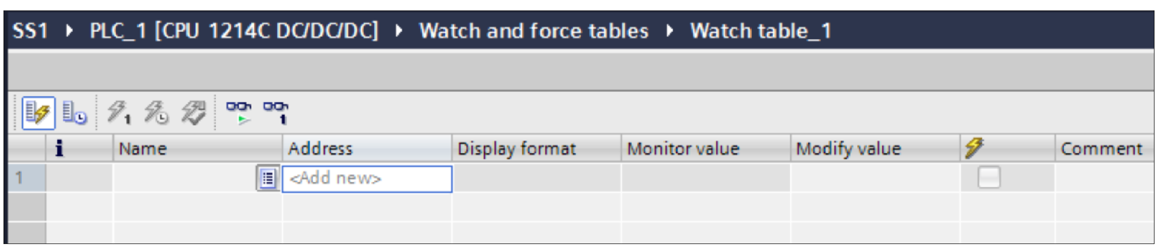
CREATE NEW WATCH WINDOW

From the **Project Tree**, select **Watch and force table** and then select **Add new watch table**.



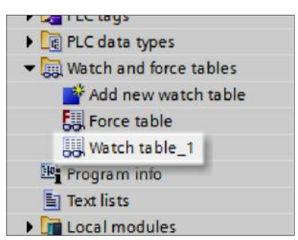
The default **Watch table_1** is created.



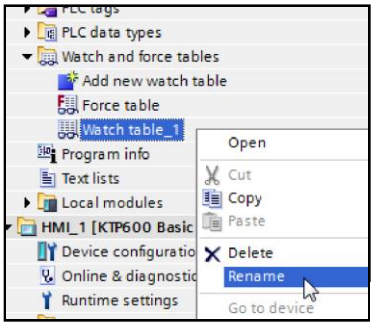


RENAME THE WATCH WINDOW

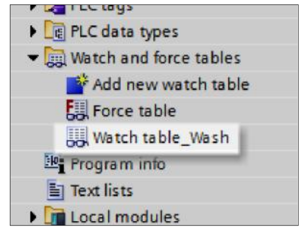
Select the Watch Window to be renamed.



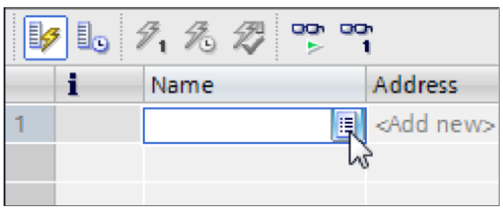
Right click and select **Rename**.



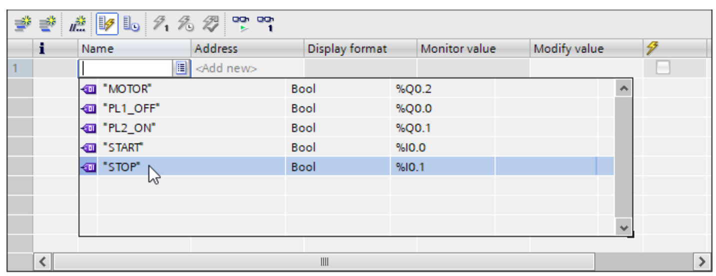
Type in new name



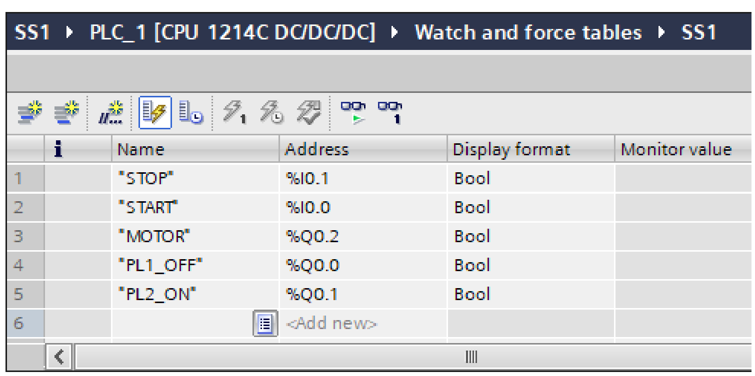
ADD TAGS TO WATCH WINDOW



To select tags, left click on **List** icon the **Name** field dropdown arrow and select the tag from the listing.



Completed Quick Watch.

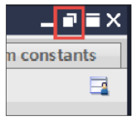


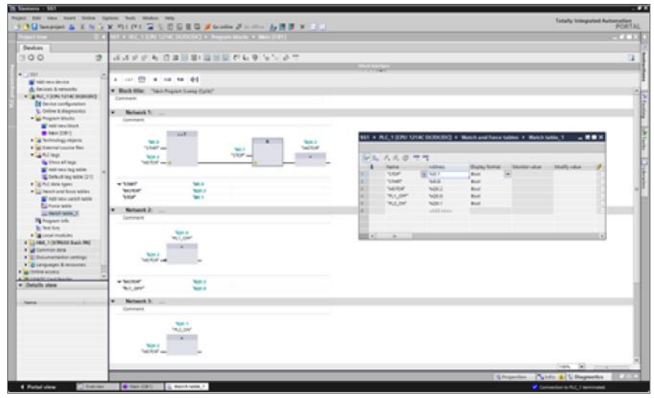
NOTE: If the Project is ***not saved***, the Watch Window will be lost.

continue

FLOAT WATCH WINDOW

Select the Float icon in the watch Window.





The Watch Window can then be resized and positioned for best viewing of multiple screens.

MONITOR WATCH WINDOW

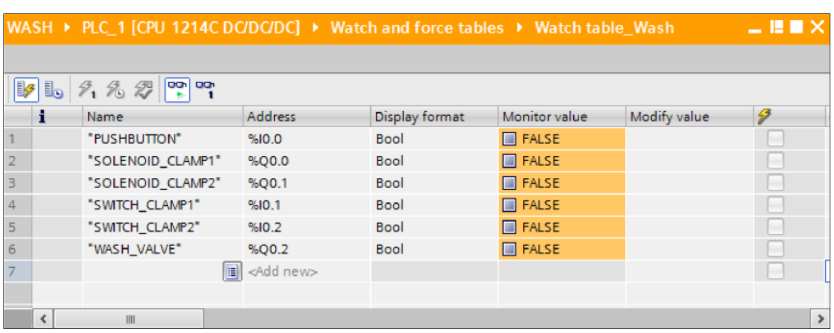
Do this step after the Project has been Downloaded to PLC.

In the Online Watch Window, select the **Monitor** icon.

WASH ▸ PLC_1 [CPU 1214C DC/DC/DC] ▸ Watch and force tables ▸ Watch table_Wash

	i	Name	Address	Display format	Monitor value	Modify value	
1		"PUSHBUTTON"	%I0.0	Bool	FALSE		
2		"SOLENOID_CLAMP1"	%Q0.0	Bool	FALSE		
3		"SOLENOID_CLAMP2"	%Q0.1	Bool	FALSE		
4		"SWITCH_CLAMP1"	%I0.1	Bool	FALSE		
5		"SWITCH_CLAMP2"	%I0.2	Bool	FALSE		
6		"WASH_VALVE"	%Q0.2	Bool	FALSE		
7			<Add new>				

The Monitored Watch Window shows the Boolean values as True or False with a small square as Green when True.



DOWNLOAD

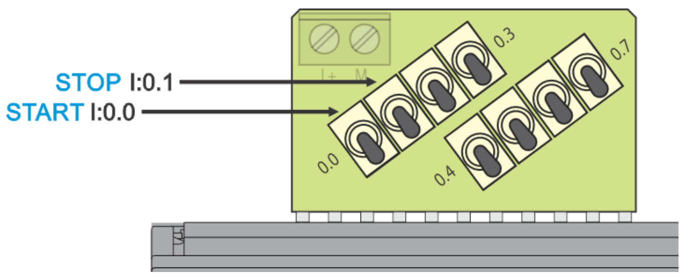
Download the program to the PLC.

There is no HMI.

Select the **Monitor** icon for the PLC program. 

In the Online Watch Window, select the **Monitor** icon. 

TESTING



Initial Conditions.

- Turn the switch **I0.1** STOP to the **up** position.
- The STOP Monitor value is set to **TRUE**.
- Turn the switch **I0.0** START to the **down** position.
- The START Monitor value is set to **FALSE**.
- The PL1_OFF indicator is On; the Monitor value is set to **TRUE**.
- The PL2_ON indicator is Off, the Monitor value is reset to **FALSE**.

Pressing and Releasing the START pushbutton.

- Turn the switch **I0.0** START to the **up** position.
- The START Monitor value is set to **TRUE**.
- Turn the switch **I0.0** START to the **down** position.
- The START Monitor value is set to **FALSE**.
- MOTOR will turn On, the Monitor value is set to **TRUE**.
- The PL1_OFF indicator is Off, the Monitor value is reset to **FALSE**.
- The PL2_ON indicator is On, the Monitor value is set to **TRUE**.

Pressing the STOP pushbutton.

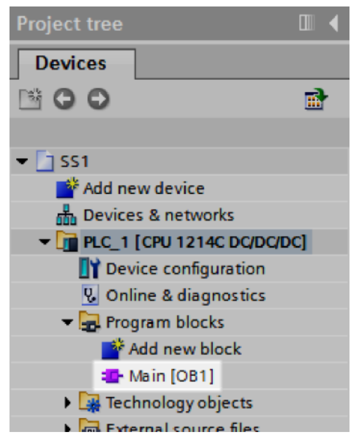
- Turn the **switch I0.1** STOP to the **down** position.
- The STOP Monitor value is set to **FALSE**.
- MOTOR will turn Off, the Monitor value is reset to **FALSE**.
- The PL1_OFF indicator is On; the Monitor value is set to **TRUE**.
- The PL2_ON indicator is Off, the Monitor value is reset to **FALSE**.

LAB: WASHER STATION TIMER **PROCESS MODEL:** WASH_TON *{last name} {first name}* **OBJECTIVE:** Create function block program using the timer on instruction. **SOFTWARE:** RSLogix5000 and RSEmulate5000

Create a program that when a momentary normally open pushbutton is pressed and held, two clamp solenoids will be energized. A normally open limit switch for each clamp will be set when that clamp is in position. When the two clamps are in position, a spray wash will clean the part for ten seconds. Releasing the pushbutton will turn off the spray wash and clamps.

PROGRAM LOCATION:

Siemens Folder
Course Day/Time Folder
Languages Folder
Function Block Folder



FOLDER\FILE NAME: WASH_TON

PROGRAM BLOCK:

Expand the Project Tree and select the routine **Main (OB1)**.

The blank editor will be displayed.

PLC Tags

INPUT:

Address	PLC IO Tag	Description
I0.0	PUSHBUTTON	Pushbutton Wash Cycle
I0.1	SWITCH_CLAMP1	Limit Switch Clamp 1
I0.2	SWITCH_CLAMP2	Limit Switch Clamp 2

OUTPUT:

Address	PLC IO Tag	Description
Q0.0	SOLENOID_CLAMP1	Clamp 1 Solenoid
Q0.1	SOLENOID_CLAMP2	Clamp 2 Solenoid
Q0.2	WASH_VALVE	Spray Wash Valve

FUNCTION BLOCK ROUTINE

See Function Block Assist for program.

Add the following instructions.

Instruction Folder for **Bit logic operations**. Bit logic operations

>=1 OR logic operation

& AND logic operation

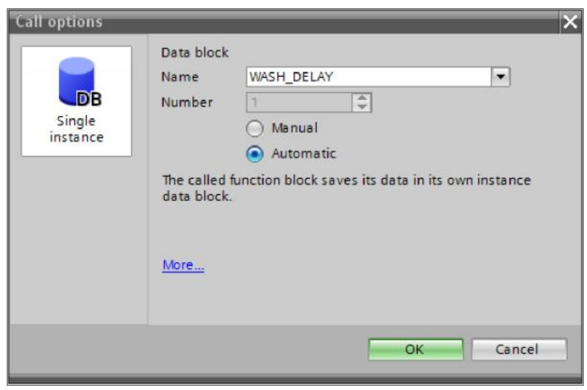
-[=] Assignment

Instruction Folder for **Bit logic operations** Timer operations

TON Timer ON

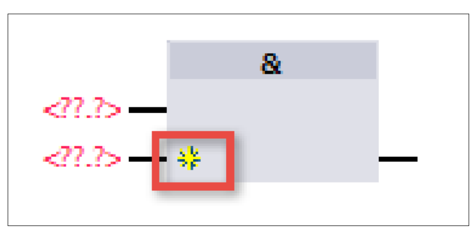
After adding theTimer ON instruction the timer's Data block dialog appears.

Change the **Name** to WASH_DELAY.



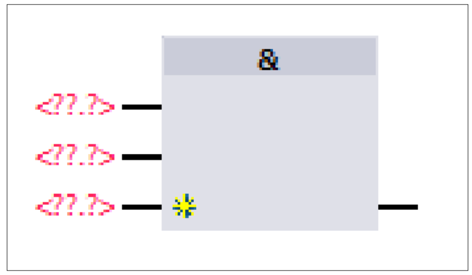
ADDING An In Pin

Left click on the Insert Input icon.



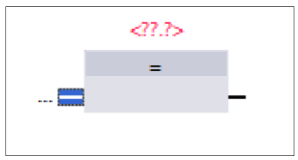
continue

Additional In pin is added.

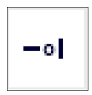


NOT - Normally Closed

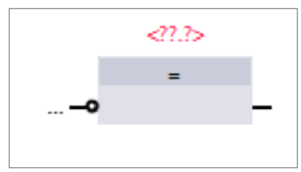
Select the Input.



Select the Invert RLO icon.



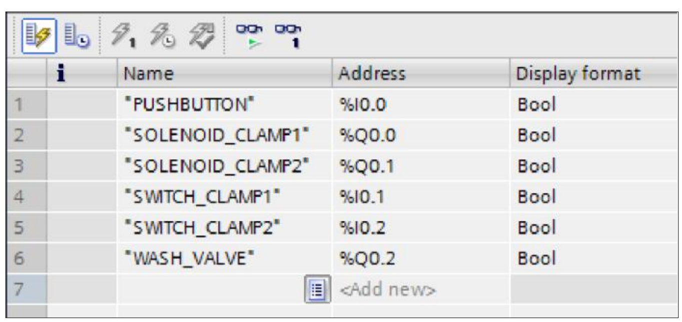
The *Results of Logic Operations* is equivalent of a Normally Closed input.



WATCH WINDOW

See SS1 for similar details on Watch Window creation and usage, pages 3 to 5.

Completed Quick Watch.



	i	Name	Address	Display format
1		"PUSHBUTTON"	%I0.0	Bool
2		"SOLENOID_CLAMP1"	%Q0.0	Bool
3		"SOLENOID_CLAMP2"	%Q0.1	Bool
4		"SWITCH_CLAMP1"	%I0.1	Bool
5		"SWITCH_CLAMP2"	%I0.2	Bool
6		"WASH_VALVE"	%Q0.2	Bool
7			<Add new>	

NOTE: If the Project is *not saved*, the Watch Window will be lost.

DOWNLOAD

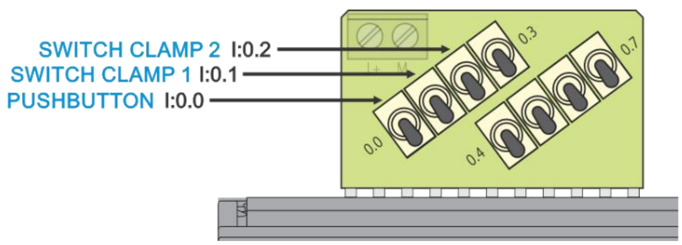
Download the program to the PLC.

There is no HMI.

Select the **Monitor** icon for the PLC program. 

In the Online Watch Window, select the **Monitor** icon. 

TESTING



Pressing the PUSHBUTTON.

- Turn the switch **I0.0** PUSHBUTTON to the **up** position.
- The PUSHBUTTON Monitor value is set to **TRUE**.
- The SOLENOID_CLAMP1 will turn On, the Monitor values are set to **TRUE**.
- The SOLENOID_CLAMP2 will turn On, the Monitor values are set to **TRUE**.
- Turn the switch **I0.1** SWITCH_CLAMP1 to the **up** position.
- The SWITCH_CLAMP1 turns On, the Monitor value is set to **TRUE**.
- Turn the switch **I0.2** SWITCH_CLAMP2 to the **up** position.
- The SWITCH_CLAMP2 turns On, the Monitor value is set to **TRUE**.
- The WASH_VALVE turns On, the Monitor value is set to **TRUE**.

- WASH_DELAY elapsing increments.
- After 10 Second, the WASH_VALVE turns Off, the Monitor value is **FALSE**.

Releasing the PUSHBUTTON.

- Turn the switch **I0.0** PUSHBUTTON to the **down** position.
- Changes PUSHBUTTON Monitor value, resets to **FALSE**.
- Turns Off the SOLENOIDS_CLAMP1, the Monitor values are reset to **FALSE**.

- Turns Off the SOLENOIDS_CLAMP2, the Monitor values are reset to **FALSE**.
- Turn the switch **I0.1** SWITCH_CLAMP1 to the **down** position.
- The SWITCH_CLAMP1 turns Off, the Monitor value is reset to **FALSE**.
- Turn the switch **I0.2** SWITCH_CLAMP2 to the **down** position
- The SWITCH_CLAMP2 turns Off, the Monitor value is set to **FALSE**.

LAB: TIMED PROCESS OUTPUT
PROGRAM FILE: PROCESS_OUTPUT *{last name} {first name}*
OBJECTIVE: Function and arithmetic instructions.

Create the program for a process that uses multi-layers of a sheet material. The type and number of sheets loaded depends on the order. The operator enters one of four types of sheet materials that will be used and the number of sheet layers. Only one type of material is used at a time, and up to six sheets may be used.

The operator presses a HMI panel pushbutton that corresponds to the material type, then presses a HMI panel pushbutton for the number of sheets. When the operator presses and releases the START pushbutton, a process output will turn on for the computing determined time value based on the type and number of sheets. The following parameters are used to then control the process timer.

Material A - multiplier of 0.73 times the number of sheets
Material B - multiplier of 1.05 times the number of sheets
Material C - multiplier of 1.23 times the number of sheets
Material D - multiplier of 1.86 times the number of sheets

PROGRAM LOCATION:

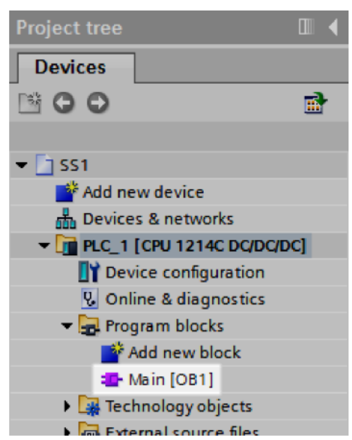
Siemens Folder
Course Day/Time Folder
Languages Folder
Function Block Folder

FOLDER\FILE NAME: PROCESS_OUTPUT

PROGRAM BLOCK:

Expand the Project Tree and select the routine **Main (OB1)**.

The blank editor will be displayed.



PLC Tags

INPUT:

Address	PLC I/O Tag	Comments
I.0.0	SW_START	START Momentary N. O. Pushbutton

OUTPUT:

Address	PLC I/O Tag	Comments
Q0.0	PROCESS_OUT	Process Output

Create the Global Data Blocks and rename. Add the following Tags.

Name of Data Block: Data_block_Process

DATA BLOCK INTERNALS:

Data Block Tag	Data Type	Comments
A_MATERIAL_MULTIPLIER	Real	Material A Multiplier
B_MATERIAL_MULTIPLIER	Real	Material B Multiplier
C_MATERIAL_MULTIPLIER	Real	Material C Multiplier
D_MATERIAL_MULTIPLIER	Real	Material D Multiplier
MATERIAL_MULTI_VALUE	Real	Material Multiplier Value
PROCESS_VALUE	Real	Process Value

Name of Data Block: Data_block_HMI

DATA BLOCK INTERNALS:

Data Block Tag	Data Type	Comments
HMI_TYPE_SHEET	Integer	Operator Panel Pushbuttons Value for Sheet Type
HMI_NUM_SHEET	Integer	Operator Panel Pushbuttons Value for Number of Sheets
HMI_PT_PROCESS_TIME	Real	Operator Panel Process Preset Time Value for program Timer Preset.
HMI_ET_PROCESS_TIME	Real	Operator Panel Process Elapsed Time Value for Program Timer Elapsed.
HMI_SW_START	Boolean	HMI Start Switch

FUNCTION BLOCK ROUTINE

See Function Block Assist for program.

The following instructions are used.

Instruction Folder for **Bit logic operations** ▾ Bit logic operations

>=1 OR logic operation

& AND logic operation

–[=] Assignment

Instruction Folder for **Bit logic operations** ▾ Timer operations

TON Timer ON

Instruction Folder for **Math Functions** ▾ Math functions

MUL Multiply

DIV Divide

Instruction Folder for **Move operations** ▾ Move operations

MOVE Move

Instruction Folder for **Word logic operations** ▾ Word logic operations

MUX Multiplex

Instruction Folder for **Conversion operations** ▾ Conversion operations

CONVERT Convert value (Data type)

Extended Instructions, Instruction Folder for **Date and time-of-day** ▾ Date and time-of-day

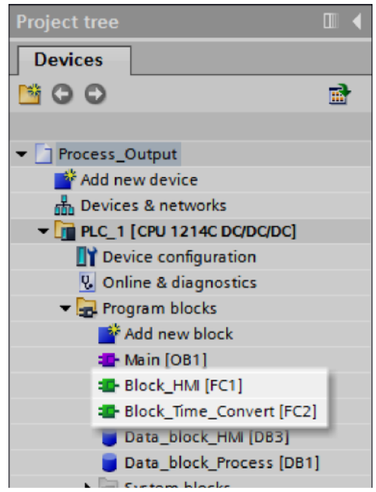
T_CONV Convert time and extract

The two program **Functions** are to be added:

Block_HMI

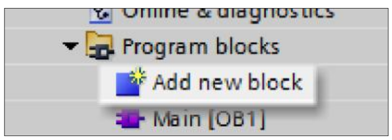
Block_Time_Convert

See below for adding a program requirements.



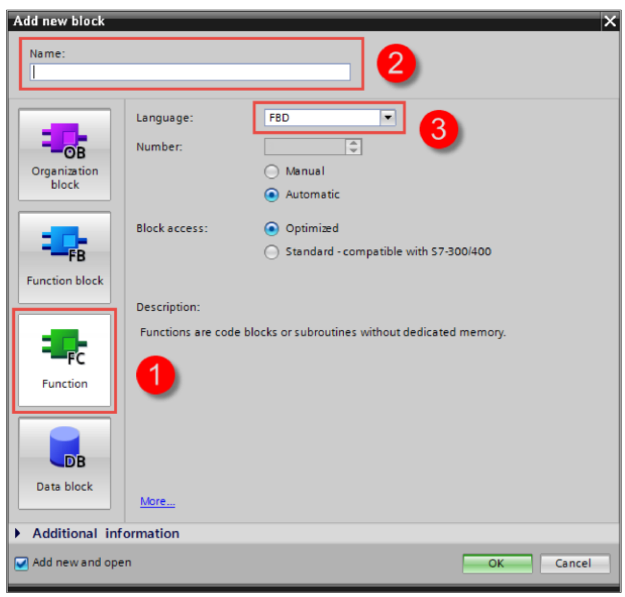
To create Function routines:

In the Project Tree select **Add new block**.



In the **Add new block** dialog.

1. Select **Function**.
2. Type in the function's **Name**.
3. Verify **FBD** is selected.
4. Select the **OK** button.



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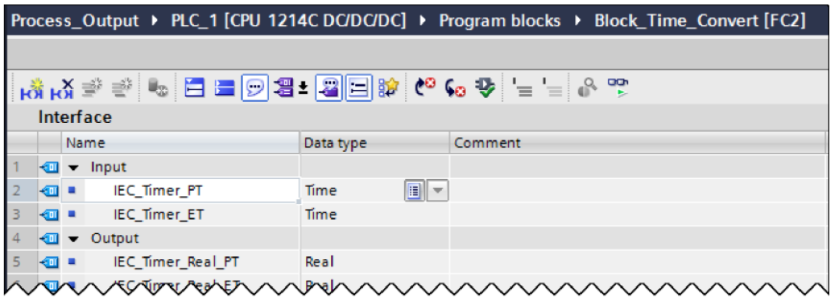
continue

Function Routine: Block Time_Convert

Add to the Interface the Input, Output, and Temp and tags.



INTERFACE for FUNCTION Block_Time_Convert:

Input Tag	Data Type	Comments
IEC_TIMER_ PT	Time	Timer PT Converted to Integer
IEC_TIMER_ET	Time	Timer PT Converted to Real

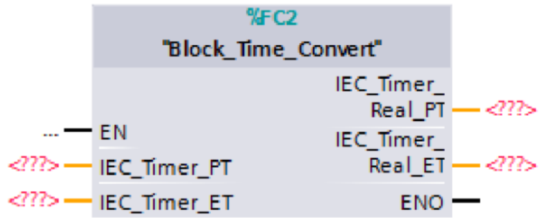
Output Tag	Data Type	Comments
IEC_TIMER_REAL_PT	Real	Timer ET Converted to Integer
IEC_TIMER_REAL_ET	Real	Timer ET Converted to Real

Temp Tag	Data Type	Comments
PT_DINT	Double Integer	Preset Value as Integer
PT_REAL	Real	Preset Value as Real
ET_DINT	Double Integer	Elapsed Value as Integer
ET_REAL	Real	Elapsed Value as Real

Time Convert extracts the Preset Time value and Elapsed Time value. Convert the double integers to real values and divide for display in seconds.

Function Routine: Block_HMI

Use the Multiplex instruction to select the process value based on the Sheet Type button pressed on the HMI to the `Material Multiplier Value`.
Drag in the Function “Block_Time_Convert” and assign the parameters.



Main [OB1]

Drag in the Function "Block_HMI".

Create program when pressing of either the hardwired or HMI Start pushbuttons tocalculate and control process output as described above. Use a multiply instruction for the product of the `Number of Material Sheets` and `Material Multiplier Value` for the `Process_Value`. A second multiply instruction converts the Process Value to milliseconds. A Convert instruction changes the millisecond `Process_Value` from a Real Data Type to Double Integer Data Type for the timer's Preset Time.

The hardwired or HMI Start pushbuttons enegizes the `Process Output` for the calculated time.

WATCH WINDOW

See SS1 for similar details on Watch Window creation and usage, pages 3 to 5.

Completed Quick Watch.

	i	Name	Address	Display format
1		"SW_START"	%I0.0	Bool
2		"Data_block_HMI".HMI_NUM_SHEET		DEC+/-
3		"Data_block_HMI".HMI_TYPE_SHEET		DEC
4		"Data_block_Process".A_MATERIAL_MULTIPLIER		Floating-point nu...
5		"Data_block_Process".MATERIAL_MULTI_VALUE		Floating-point nu...
6		"Data_block_Process".PROCESS_VALUE		Floating-point nu...
7		"Data_block_Process".PROCESS_PT		DEC+/-
8		"PROCESS_OUT"	%Q0.0	Bool
9			<Add new>	

NOTE: If the Project is *not saved*, the Watch Window will be lost.

continue

DOWNLOAD

Download the program to the PLC.

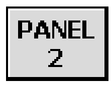
Download the **HMI** program.

Select the **Monitor** icon for the PLC program. 

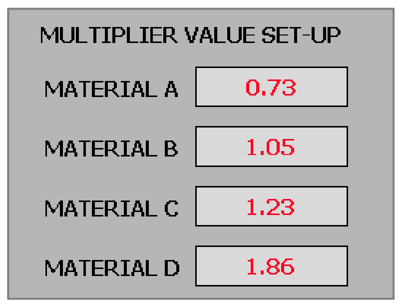
*For each opened Program the **Monitor** icon **must be** selected*

In the Online Watch Window, select the **Monitor** icon. 

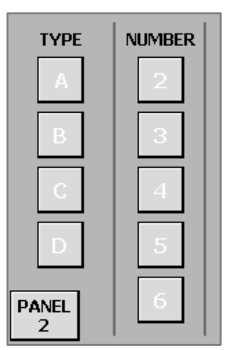
TESTING

Press the button to go to **PANEL 2**. 

Enter the values for the **A**, **B**, **C**, and **D** **MATERIALS**.



Press the button to return to the **MAIN** screen. 

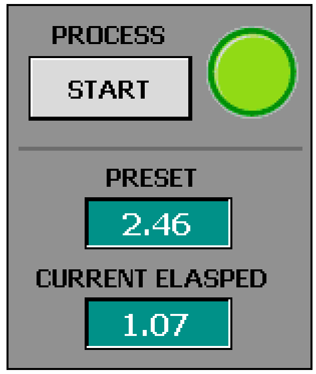


Press the Button for **TYPE B**.

Press the Button for **NUMBER 2**.

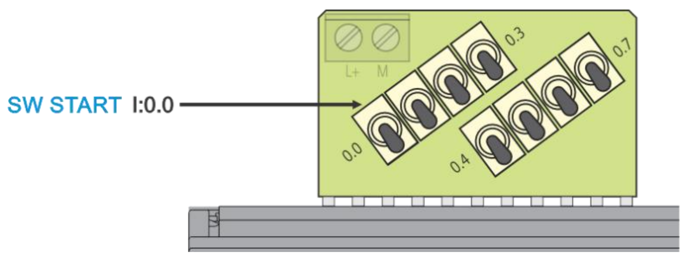
Press the **START** button.

The **OUTPUT** should be ON while the timer is **ELASPING**.



Select the other **TYPE** and **NUMBER** buttons and **START** to verify the different values calculated values for the Timer's **PRESET**.

TESTING THE START SWITCH



Push up a **switch I0.0** SW START.

The **OUTPUT** should be ON while the timer is **ELASPING**

Push down a **switch I0.0** SW START.

INFORMATION ONLY

In Block_HMI, notice for the Multiplex the **K** input uses the values 0 thru 3 for the Sheet Types instead of 1 thru 4. This is due to the Multiplex **IN**s starting with 0.

If the HMI button values where kept at 1 through 4, a tag NULL would be assigned to IN0.